

UPDATE BRIEF FOR AN ORDER ON THE LEECH LATTICE FROM A $\mathbb{Z}_3$ -SYMMETRIC TRIPLE-OCTONION PRODUCT

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ABSTRACT. The manuscript version of 29 April 2026 (v3) is frozen while under review. This brief records two updates for a future revision: one correction that *needs* to be made (a non-load-bearing error in a footnote of Section 2.3), and one simplification that *should* be made (replacing the three appendix structure-constant tables by a single one, with the other two lemmas obtained from a reduction modulo $2L$). Section and lemma numbers refer to v3. Every claim below was checked in an independent exact-arithmetic implementation.

1. REQUIRED CORRECTION: THE SECTION 2.3 FOOTNOTE

Location. The footnote attached to the sentence “Both $L\bar{s}$ and Ls are sublattices of L ” in Section 2.3. The error. The footnote states two properties of $L\bar{s}$ — that it is a $\mathbb{Z}$ -subgroup of L , and that it is “closed under left-multiplication by L ” — and then asserts: “*For Ls both properties are immediate from $s \in L$.*” The second property is **false for Ls** : a basis-level check shows that 40 of the 64 products $L_i \cdot (L_j s)$ leave Ls , so $L \cdot Ls \not\subseteq Ls$.

The claim is moreover self-contradictory within the paper. Since $Ls \subseteq L$, a closure $L \cdot Ls \subseteq Ls$ would force $Ls \cdot Ls \subseteq L \cdot Ls \subseteq Ls$ — exactly the closure that Remark 4.5 (and the very existence of Table A.3) denies. The non-closure of Ls is the central obstruction the paper resolves; the footnote inadvertently asserts it away.

Why it is not load-bearing. The footnote only needs to justify the sentence it annotates, namely that Ls is a *sublattice* of L . That requires only $Ls \subseteq L$ and that Ls is a subgroup — both genuinely immediate from $s \in L$, since $\ell s \in L \cdot L \subseteq L$. Closure under left-multiplication by L is never used for Ls anywhere in the proof; it was introduced because it is correct and relevant for $L\bar{s}$ (it is Wilson’s condition (2)) and then over-extended to Ls by the phrase “both properties.” No theorem, lemma, or proposition depends on the false claim.

Suggested fix. Replace “For Ls both properties are immediate from $s \in L$.” by, e.g.:

For Ls , the subgroup property is likewise immediate from $s \in L$. Closure under left-multiplication by L , by contrast, holds for $L\bar{s}$ but *fails* for Ls — and that failure is precisely the obstruction resolved by the twist (Remark 4.5).

This costs one sentence, removes the contradiction, and foreshadows the crux of the proof.

Remark. A related fact, correct and optional: $L\bar{s}$ is in fact a *two-sided* ideal of L ($L \cdot L\bar{s}$, $L\bar{s} \cdot L$, and $L\bar{s} \cdot L\bar{s}$ are all contained in $L\bar{s}$). The paper claims, and needs, only the left-multiplication half; no change is required.

2. RECOMMENDED SIMPLIFICATION: THREE TABLES TO ONE

Appendix A currently carries three 64-row tables of integer structure constants: Table A.1 for Lemma 4.2 ($L \cdot L \subseteq L$, textbook, cited to Coxeter), and Tables A.2 and A.3 for Lemmas 4.3 and 4.4. Lemmas 4.3 and 4.4 can instead be obtained from Table A.1 by a reduction modulo $2L$, eliminating the two extra integer tables.

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The reduction. The paper already states (Section 2.3, after Wilson) that $L\bar{s} \cap Ls = 2L$. Hence $2L \subseteq Ls$ and $2L \subseteq L\bar{s}$, and applying the involution σ (with $\sigma(2L) = 2L$),

$$2L \subseteq \sigma(Ls), \quad 2L \subseteq \sigma(L\bar{s}).$$

Because $L \cdot L \subseteq L$ (Lemma 4.2) and the octonion product is $\mathbb{Z}$ -bilinear, $(a+2c) \cdot b \equiv a \cdot b \pmod{2L}$ for $c \in L$, and likewise in the right factor. The product therefore descends to a well-defined $\mathbb{F}_2$ -bilinear product

$$\bar{\mu} : \bar{L} \times \bar{L} \rightarrow \bar{L}, \quad \bar{L} := L/2L \cong \mathbb{F}_2^8,$$

whose structure constants are exactly the entries of Table A.1 *reduced modulo 2*. No new table is introduced.

Since $2L$ is contained in both $\sigma(L\bar{s})$ and $\sigma(Ls)$, these sublattices are full preimages of $\mathbb{F}_2$ -subspaces

$$V := \sigma(L\bar{s})/2L, \quad W := \sigma(Ls)/2L,$$

and from $\sigma(L\bar{s}) + \sigma(Ls) = L$, $\sigma(L\bar{s}) \cap \sigma(Ls) = 2L$ one gets $\bar{L} = V \oplus W$ with $\dim V = \dim W = 4$. The two lemmas as subspace conditions. The closure statements become conditions on $\bar{\mu}$:

Lemma 4.3 $\iff \bar{\mu}(\bar{L}, V) \subseteq V$ (V a left ideal), Lemma 4.4 $\iff \bar{\mu}(W, W) \subseteq W$ (W a subalgebra).

The substantive direction is immediate: for Lemma 4.3, given $a \in L$ and $w \in \sigma(L\bar{s})$, the product $a \cdot w$ lies in L (Lemma 4.2) with class $\bar{\mu}([a], [w]) \in V$; since $\sigma(L\bar{s})$ is the full preimage of V , this gives $a \cdot w \in \sigma(L\bar{s})$. Lemma 4.4 is identical with W . Both reduced conditions have been verified.

Effect on the appendix. Tables A.2 and A.3 (128 rows of unbounded integers) are replaced by checks against a single object — the mod-2 reduction of Table A.1 — together with the mod-2 generators of V and W (the appendix's N_k and M_k reduced modulo 2). The two checks are $\bar{\mu}(\bar{L}, V) \subseteq V$ ($8 \times 4 = 32$ products over $\mathbb{F}_2$) and $\bar{\mu}(W, W) \subseteq W$ ($4 \times 4 = 16$ products), small enough to display compactly or inline. Table A.1 is kept as the worked demonstration of method and becomes the single integer table carrying all three lemmas.

A structural by-product. The Conclusion lists as open “a structural reason for Lemma 4.4.” The reduction supplies a partial answer in the requested register: closure of $\sigma(Ls)$ is the statement that the 4-dimensional subspace W is a subalgebra of the 8-dimensional $\mathbb{F}_2$ -algebra $L/2L$, and the twist's role is to move the condition-(3) subspace from $Ls/2L$ (not a subalgebra) to W (a subalgebra).

Scope and caveats.

- This is a reduction in size and a gain in transparency, not a removal of all computation. Lemmas 4.3 and 4.4 remain genuinely independent finite checks — W being a subalgebra does not follow formally from V being an ideal. What changes is that the checks are small $\mathbb{F}_2$ computations against one shared table rather than two separate $\mathbb{Z}$ -valued tables.
- The reduction consumes the equality $L\bar{s} \cap Ls = 2L$, currently cited to Wilson. For a self-contained appendix this should be stated as the hypothesis the reduction uses (it is cheap to verify directly).
- *Direct framing of Lemma 4.4.* By conjugating with σ , Lemma 4.4 is equivalent to: the twisted product closes on Ls ,

$$Ls \cdot_{\sigma} Ls \subseteq Ls, \quad \text{even though} \quad Ls \cdot Ls \not\subseteq Ls.$$

This is the most direct way to *state* the crux — it is the octonion-level counterpart of the “two equivalent framings” of Remark 4.6. It is a relabelling, however, and does not by itself avoid the appendix table: a proof is still required, and it is the reduction modulo $2L$ above that supplies the table-free proof. The two are complementary — the direct framing for the statement, the modular reduction for the proof.

- Adopting this is an exposition choice. Recasting Lemmas 4.3 and 4.4 in the mod-2 language shortens the appendix and strengthens the structural narrative of the Conclusion; keeping all three integer tables remains defensible as a uniform, fully explicit presentation. The mathematics of the paper is unchanged either way.

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